

Ex 2 : Exploratory Data Analysis (EDA) & Data Visualization

Dataset: FIFA World Cup Match History (`ex2_matches.csv`)

Building on the preprocessing techniques from **Exercise 1**, this exercise focuses on conducting Exploratory Data Analysis (EDA) on real-world historical sports data. You will identify missing value patterns, handle missing values contextually, detect statistical outliers visually and programmatically, and generate meaningful visualizations.

1. Initial Dataset Inspection: Load the dataset into a pandas DataFrame, display the dimensions and data types. Calculate and display the count and percentage of Missing values of each column.
2. Visualizing & Handling Missing Data: Generate a heatmap to visualize missing data across all columns. Identify the key columns containing missing values and impute them using an appropriate strategy. Remove or fill missing values in non-numeric columns using standard techniques.
3. Compute a new column, **Total Goals per Match** where:
$$\text{Total_Goals} = (\text{Home_Team_Goals} + \text{Away_Team_Goals}).$$
4. Trend Analysis: Create a line plot showing the average goals per match across World Cup years (1930–2022). Include markers, custom gridlines, and clear axis labels.
5. Category Analysis: Create a horizontal bar chart displaying the Top 10 teams with the highest total goals scored across all World Cup matches.
6. Outlier Detection & Analysis: Create box plots for `attendance` and `Total_goals` to detect outliers. Use IQR to write a function that extracts Rows where `Attendance` or `Total_Goals` are classified as statistical outliers.
7. Sumamry: At the end of your notebook, document the insights derived from your visualizations and analysis.